UNIVERSITY OF WESTMINSTER

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University of Westminster
University Research Ethics Committee

Application for Research Ethics

PART A

Section 1 – PROJECT AND APPLICANT DETAILS

1.1 Project Title: **Usability Testing & Evaluation – Group Coursework (Google Flights)**

1.2 Applicant Details	
Name: Thalal Izzath Anika Mohamed Linushan Sivakumar Saad Mazhar	**University Email Address:** Thalal.20200165@iit.ac.lk Anika.20232502@iit.ac.lk Linushan.20232762@iit.ac.lk Saad.20221804@iit.ac.lk
Contact Address: IIT Srilanka	**Telephone Number:** +94 77 477 4789 +94 77 765 7209 +94 76 726 6624 +94 72 013 5801
Faculty: Computing Faculty	
Please check the relevant box: **Undergraduate** ☒ **Postgraduate** ☐ **MPhil/PhD Student** ☐ **Staff** ☐	
I confirm I have read the *University's Code of Practice Governing the Ethical Conduct of Research*	**YES** ☒ **NO** ☐

1.3 Supervisor/Dean of Faculty/Faculty Research Director details

Please note that all applicants with a supervisor(s) must ensure that the supervisor signs the declaration at the bottom of this page if completing Part A only or in **Section 10.3** if completing Part B

All **staff** must ensure that their Dean of Faculty, or Faculty Research Director (or nominee), as appropriate, signs the declaration at the bottom of this page if completing Part A only or in **Section 10.3** if completing Part B

Name: Mrs. Savinthie Suwandaratna	**University Email Address:** savinthie.s@iit.ac.lk
Faculty: Computing	**Telephone Number:**

PART A (Continued)

Section 2 – Project Details

2.1 Please provide a description of the background with references to relevant literature (250 words maximum):

The usability of complex web services, particularly those managing large datasets and critical consumer decisions, is paramount for user adoption and satisfaction. Platforms like Google Flights, which enable users to search, filter, and select travel itineraries, must maintain high standards of Effectiveness, Efficiency, Error Tolerance, and Ease of Learning. Usability testing is the established methodological approach used to assess how effectively users interact with such systems.

This study aims to evaluate the Google Flights platform across these key usability dimensions to identify specific pain points and opportunities for improvement. The chosen methodology involves designing and conducting a structured usability test to capture both objective quantitative data and subjective qualitative data. This approach is crucial for establishing an evidence-based understanding of the user experience.

2.2. Please provide a brief description and the aims of your study (250 words maximum):

This research study aims to assess the usability and user experience of the Google Flights platform, focusing on its core features for searching and applying advanced travel filters. As Google Flights is a critical tool for consumer travel planning, its interface must facilitate efficient, error-free, and satisfying user journeys. The objective is to evaluate the platform against established usability metrics: Effectiveness, Efficiency, Error Tolerance, and specifically, Ease of Learning. The study employs structured usability testing with a mixed-experience user group. Data collection is implemented via a Within-Subjects Design to ensure robust comparisons, capturing both quantitative and subjective data across specific tasks involving complex filter application.

The primary aims of this study are to identify potential usability problems users encounter when utilizing Google Flights' advanced filtering tools and to formulate evidence-based recommendations for design improvements. This analysis will address various usability aspects through four distinct hypotheses, including the use of Pearson Correlation for learnability and Independent/Paired sample-tests for efficiency and effectiveness. Ultimately, the study seeks to ensure that the Google Flights interface aligns with the expectations and needs of end-users across varying levels of technical and travel-booking experience, thereby enhancing user workflow, reducing errors, and increasing overall user satisfaction with the flight booking process.

2.3. Please outline the design and methodology of your study (include details of the selection and recruitment of participants (if any) and details of any invasive (e.g. blood samples, inhalation/ingestion of food and/or non-food products (in abnormally higher or lower levels than normal or a different form), or intrusive (e.g. questionnaires, focus groups, interviews, etc.) procedures [attach extra information as necessary] (400 words maximum in total):

This study adopts a mixed-method usability testing approach to evaluate the Google Flights platform, specifically assessing effectiveness, efficiency, error tolerance, and learnability regarding its search and filtering capabilities.
The research integrates four statistical designs - Independent-sample, Paired-sample, ANOVA, and Pearson Correlation to analyze user performance across distinct flight-planning scenarios, such as multi-city bookings and complex filter usage.

The study targets young professionals, recruiting a diverse mix of novice, intermediate, and

expert users through university networks and social media channels. Data collection involves remote sessions where participants perform tasks under moderated and unmoderated conditions.

Key metrics including task success rates, completion times, and filter usage counts are recorded using screen recording software (OBS Studio) and digital timers, while subjective satisfaction and confidence are captured via pre- and post-test Google Form questionnaires. Participation is entirely voluntary with the right to withdraw at any time; the study involves no invasive procedures, and intrusive measures are limited to standard screen monitoring and surveys, ensuring all data remains anonymized and securely stored.

2.4. Timescales

Start Date (DD/MM/YY): 05/11/2025

Estimated duration of work: 15 Days

Section 3 - RISK OF HARM

NOTE 1: Where indicated below applicants should check if the research will require ethical approval from a National Research Ethics Committee via the Integrated Research Application System (IRAS) - nres.queries@nhs.net.- http://www.hra-decisiontools.org.uk/ethics/
NOTE 2: The University of Westminster holds a Human Tissue Authority Licence – This licence is specifically for tissue stored at 115 New Cavendish Street in accordance with the terms of the licence – Advice must be obtained from the University Human Tissue Designated Individual (N.Presneau@westminster.ac.uk)

	RISK OF HARM (to self, colleagues, participants, environment or animals)	Yes	No	N/A
1	Will any pain or more than mild discomfort result from the study?	☐	☒	☐
2	Could the study induce any psychological stress or anxiety or cause harm or negative consequences beyond the risks encountered in normal life?	☐	☒	☐
3	Will the study involve prolonged or repetitive physical or psychological testing of human participants that may put someone at risk, e.g. use of treadmill?	☐	☒	☐
4	Will the study involve raising sensitive topics (e.g. sexual activity, drug use, revelation of medical history, bereavement, illegal activities, etc.)?	☐	☒	☐
5	Does your work involve any "relevant material" containing human cells (e.g. blood, urine, saliva, body tissues but NOT established cell-lines) from living or deceased persons (Such work must take account of the Human Tissue Act)? – See Note 1 and 2 above.	☐	☒	☐
6	Will DNA samples be taken from human participants (Such work must take account of the Human Tissue Act)? – See Note 1 and 2 above.	☐	☒	☐
7	Does your study raise any issues of personal safety for you or other researchers or participants involved in the project (Especially relevant if taking place outside working hours or off University premises)?	☐	☒	☐
8	Does your study involve deliberately misleading the participants (e.g. deception, covert observation)?	☐	☒	☐
9	Does your work involve administration of a food or non-food substance of a different type from or in abnormally higher or lower amounts than normal or one that is known to cause allergic reaction(s) or potential psychological stress?	☐	☒	☐
10	Does your study involve issues relating to personal and/or sensitive data?	☐	☒	☐
	PARTICIPANTS (and/or their records/associated data) Does your work involve any of the following:	**Yes**	**No**	**N/A**
11	Human participants in a health and/or social care setting (e.g. patients, those attending day centres, community care, rehabilitation centres, etc., including in the NHS, other public, private and/or voluntary sectors)? – See Note 1 above.	☐	☒	☐
12	Human participants who may be deemed vulnerable (e.g. children, people in poverty and/or with physiological or psychological impairments, persons attending rehabilitation centres, persons in easily identifiable positions that could be subject to victimisation, etc.)?	☐	☒	☐
13	Expectant or new mothers?	☐	☒	☐
14	Refugees/Asylum seekers?	☐	☒	☐
15	Minors (under the age of 18 years old)?	☐	☒	☐
16	Participants in custody (e.g. prisoners or arrestees)? – See Note 1 above.	☐	☒	☐
17	Participants with impaired mental capacity (e.g. severe mental illness, brain damage, sectioned under Mental Health Act, lowered or reduced sense of consciousness)? – See Note 1 above.	☐	☒	☐
18	Animals (or animal tissue).	☐	☒	☐
	INFORMATION TO PARTICIPANTS	**Yes**	**No**	**N/A**
19	Will you provide participants with a Participant Information Sheet prior to obtaining informed consent which can be taken away by the participant?	☒	☐	☐
20	Will you describe the procedures to participants in advance, so that they are informed about what to expect?	☒	☐	☐
21	Will you obtain informed consent for participation (normally written)? OR in the case of using personal data previously acquired was consent given for the reuse of the data for other research purposes?	☒	☐	☐
22	Will you tell participants that they may withdraw from the research at any time and for any reason without any impact on their care, service provision etc.?	☒	☐	☐
23	Will you give participants the option of omitting questions they do not want to answer?	☒	☐	☐
24	Will you tell participants that their data will be treated as confidential and that, if published, it will not be identifiable as theirs?	☒	☐	☐
25	Will you offer feedback to participants at the end of their participation, upon request (e.g. give them a brief explanation of the study and its outcomes)?	☒	☐	☐
26	Has external funding or collaboration been applied for/received, which requires	☐	☐	☒

	institutional ethical consideration or approval?			

Useful links:

- http://www.screc.org.uk/ - Social Care Research Ethics Committee
- http://www.hra-decisiontools.org.uk/ethics/ - Human Research Authority decision tool to identify if research needs National Research Ethics Committee approval
- http://www.nres.nhs.uk/applications/guidance/governance-and-directives/?entryid62=131341 – Governance Arrangements for Research Ethics Committees
- http://www.nres.nhs.uk/EasySiteWeb/GatewayLink.aspx?alId=134016 - NRES algorithm "Does my project require review by a Research Ethics Committee"?
- http://www.hta.gov.uk/policiesandcodesofpractice/codesofpractice.cfm - Human Tissue Authority Code of Practice
- http://www.hta.gov.uk – Human Tissue Authority website
- http://www.rsclearn.mrc.ac.uk/MRC_HumanTissueAct/player.html - Medical Research Council online training course for Human Tissue Act.

What to do next:

- **If you have answered NO to questions 1-18 (inclusive) and YES to questions 19-25 (inclusive)**, you do not need to complete the Full Research Ethics Approval Form (Part B). Please keep this form for your records, and do not submit to **Faculty Research Ethics Committee** (FREC) unless you require ethical consideration of your study, regardless of ethical implications, by an external body (question 26 has been answered YES). **A list of Faculty contacts is below.**
- **If you have answered YES to any of the questions 1-18 (inclusive) or NO to any of the questions 19-25** the Full Research Ethics Approval Form (Part B) MUST be submitted including Cover Sheet, Part A and Part B of the application form plus any required supplementary documents to the Secretary of the relevant Faculty Research Ethics Committee (FREC). **A list of Faculty contacts is below.**
- If you are applying for external Ethical Approval, please send a *copy* of the Conditions/Approvals letters to the **University Research Ethics Committee** (UREC) Secretary (this may include the original ethical application(s)). Where the external ethics committee/body has equal standing or primary jurisdiction, e.g. another University Research Ethics Committee or a National Research Ethics Committee, any approval will normally be received and noted by the University of Westminster Research Ethics Committee and further consideration may not be required. Where the external committee does not have equal or higher standing than the University Committee then the full ethical approval process at the university may still be required. Additional institutional compliance issues may need consideration by UREC.
- All Applications (dated, signed and authorised) and supplementary information or External Approvals should be sent to the University Research Ethics Committee (UREC) Secretary in *electronic format with a version number, document name and date and the Principal Investigator (or Undergraduate/Postgraduate Taught Student) name. On receipt your application will be issued a unique reference number*
- All new Applications should be submitted to a Research Ethics Committee (FREC or UREC) Secretary a minimum of 10 working days in advance of the Committee meeting date (earlier submission is recommended so that applications can be pre-vetted and obvious issues addressed before the application is considered by the Committee).

Contact details:

Faculty	Chair	Secretary
Architecture and the Built Environment	Professor Nick Bailey	Colette Davis
Media Arts and Design	Dr Anthony Mcnicholas	Fauzia Ahmad

Faculty	Chair	Secretary
Science and Technology	Dr John Colwell	Mandy Walton
Science and Technology Psychology Department Sub Committee	Dr Laura Boubert	TBC
Social Sciences and Humanity	Professor Marco Roscini	Victoria Grey-Edwards
Westminster Business School	Petar Sudar	Haydn Worley
University Research Ethics Committee	Professor Graham Megson	Huzma Kelly

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